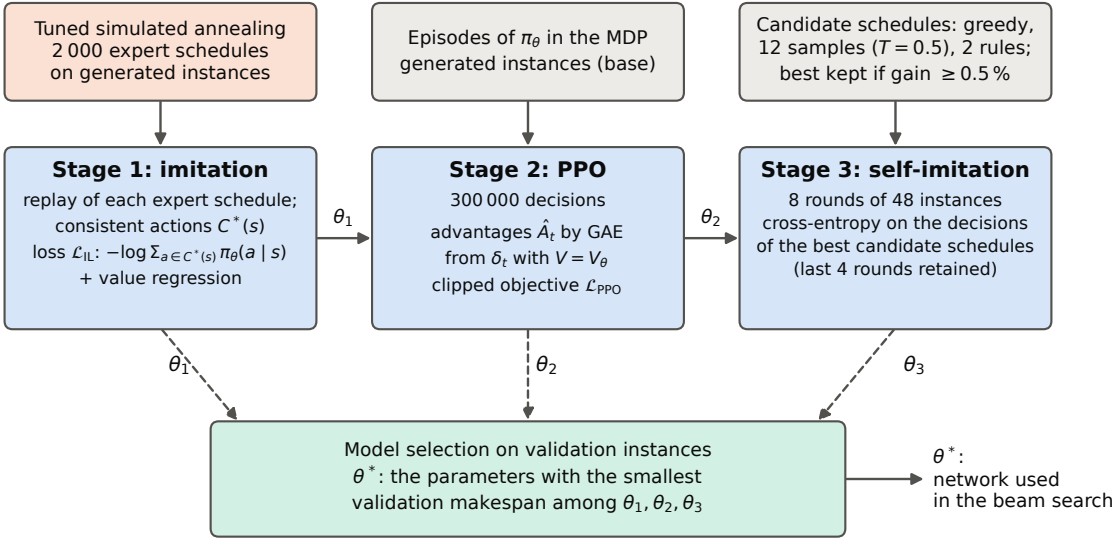
Tuned simulated annealing
2 000 expert schedules
on generated instances
Episodes of π_θ in the MDP
generated instances (base)
Candidate schedules: greedy,
12 samples ($T = 0.5$), 2 rules;
best kept if gain $\geq 0.5\,\%$
Stage 1: imitation
replay of each expert schedule;
consistent actions $C^*(s)$
loss $\mathcal{L}_{\text{IL}}$: $-\log \Sigma_{a \in C^*(s)} \pi_\theta(a \mid s)$
+ value regression
θ_1
Stage 2: PPO
300 000 decisions
advantages $\hat{A}_t$ by GAE
from δ_t with $V = V_\theta$
clipped objective $\mathcal{L}_{\text{PPO}}$
θ_2
Stage 3: self-imitation
8 rounds of 48 instances
cross-entropy on the decisions
of the best candidate schedules
(last 4 rounds retained)
θ_1
θ_2
θ_3
Model selection on validation instances
θ^*: the parameters with the smallest
validation makespan among $\theta_1, \theta_2, \theta_3$
θ^*:
network used
in the beam search